

SCHULER AGDM, AB, Canada

WMO: 712690

Lat: 50.3069N

Lon: 110.0906W

Elev: 2625

StdP: 13.35

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.1	-14.6	-26.6	1.4	-20.1	-20.6	2.0	-14.4	30.7	30.0	27.1	28.2	7.8	260	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	90.1	61.9	86.5	61.4	82.7	60.7	65.7	81.5	64.0	79.3	62.3	77.8	10.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	87.5	70.0	58.5	80.7	68.8	56.6	75.3	66.9	32.0	81.4	30.6	79.2	29.3	77.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	24.4	21.6	DB	-27.9	96.2	4.8	2.9	-31.4	98.3	-34.2	100.0	-36.8	101.7	-40.3	103.8	(4)
(5)			WB	-28.2	70.2	4.7	2.6	-31.6	72.1	-34.3	73.6	-37.0	75.1	-40.4	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	17.4	16.0	29.5	41.8	52.0	59.3	66.1	64.6	55.2	42.5	28.0	16.8		
	DBStd	21.12	15.35	14.57	14.12	8.81	8.12	5.73	5.50	6.40	8.64	9.91	13.16	14.50		
	HDD50	4947	1009	952	638	265	73	4	0	0	43	271	662	1030		
	HDD65	8991	1474	1372	1101	695	407	187	54	87	310	699	1110	1495		
	CDD50	1630	0	0	1	20	136	283	499	452	199	38	2	0		
	CDD65	199	0	0	0	0	5	15	88	74	16	1	0	0		
	CDH74	3759	0	0	0	10	178	341	1471	1328	410	21	0	0		
	CDH80	1379	0	0	0	1	34	87	575	531	149	2	0	0		
	WSAvg	10.6	12.0	10.3	10.9	11.4	11.2	10.4	8.8	8.8	9.7	10.9	11.4	11.0		
	PrecAvg	13.7	0.6	0.5	0.6	0.9	1.7	3.1	1.7	1.7	1.1	0.9	0.6	0.6		
(15) Precipitation	PrecMax	20.1	1.7	1.5	1.6	2.4	5.3	6.3	4.4	4.9	2.8	2.3	1.7	1.2		
	PrecMin	8.3	0.1	0.0	0.0	0.0	0.0	1.7	0.2	0.0	0.0	0.2	0.1	0.2		
	PrecStd	3.1	0.4	0.4	0.4	0.6	1.3	1.4	1.1	1.1	0.7	0.5	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	49.4	64.0	74.6	84.5	87.6	94.9	94.3	90.4	77.0	63.3	48.4		
		MCWB	40.2	39.6	48.0	51.3	58.2	62.9	64.5	62.0	59.6	54.8	47.3	40.9		
	2%	DB	43.0	42.9	57.7	68.7	79.0	82.0	90.5	89.9	84.5	69.6	54.9	42.8		
		MCWB	37.4	36.3	44.5	49.3	55.9	61.4	63.4	61.0	58.3	51.5	43.6	36.7		
	5%	DB	40.0	38.5	52.3	63.9	74.3	77.7	86.4	86.2	78.5	63.7	49.7	39.3		
		MCWB	35.8	33.7	42.2	46.7	53.7	59.9	63.1	60.4	56.1	49.3	40.4	34.9		
	10%	DB	37.6	34.9	46.7	58.6	69.4	73.6	82.4	81.9	72.6	58.3	45.2	36.0		
		MCWB	34.1	31.4	38.8	44.2	51.1	58.0	62.3	59.9	54.1	46.3	37.9	32.3		
	0.4%	WB	40.7	40.8	48.8	53.0	59.3	67.0	69.9	65.9	61.3	56.4	48.2	41.7		
		MCDB	46.7	47.7	62.6	70.7	78.3	81.1	84.2	82.9	83.0	73.6	59.5	48.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.8	36.7	45.2	50.2	57.1	63.6	67.0	64.0	59.1	52.8	44.6	37.5		
		MCDB	42.3	42.3	55.9	66.8	74.5	77.8	82.7	81.2	79.5	67.0	54.6	42.0		
	5%	WB	36.0	34.0	42.2	47.6	54.9	61.2	65.1	62.4	56.9	49.7	41.1	34.8		
		MCDB	39.9	38.2	51.3	61.8	71.1	74.8	80.2	79.8	75.8	61.7	48.8	39.1		
	10%	WB	33.9	31.6	39.2	45.0	52.8	59.0	63.4	60.9	54.8	47.0	38.1	32.3		
		MCDB	37.5	34.8	46.6	56.6	66.9	71.0	78.2	77.8	70.8	57.3	44.2	35.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	19.2	20.2	23.3	24.0	22.6	27.3	27.2	25.1	22.6	19.2	17.0		
		MCDBR	18.1	22.2	26.2	33.0	32.2	30.3	35.1	36.0	36.3	30.3	25.5	17.4		
		MCWBR	14.0	16.5	15.4	16.9	13.8	13.9	13.5	13.2	14.6	15.4	15.7	13.1		
	5% WB	MCDBR	17.6	20.9	25.2	30.2	28.4	27.8	29.7	30.1	33.2	29.0	24.1	17.4		
		MCWBR	13.9	15.9	15.3	16.0	12.8	14.3	13.5	12.1	14.1	15.4	15.7	13.3		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.253	0.289	0.349	0.373	0.362	0.368	0.385	0.335	0.295	0.262	0.245		
	taud		2.378	2.499	2.467	2.301	2.277	2.370	2.367	2.285	2.406	2.502	2.490	2.429		
	Ebn at Noon		250	291	294	283	280	284	280	268	272	268	253	244		
	Edh at Noon		20	23	28	37	40	37	36	38	30	23	17	16		
(44) All-Sky Solar Radiation	RadAvg		385	700	1131	1556	1854	1988	2158	1745	1259	776	428	297		
	RadStd		40	62	78	109	124	125	114	119	105	82	36	28		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page